Input Height Bounds and Odd Contact Radicals

10 November 2026

Abstract

A positive power of a rational input gives a height scale whose growth is uniform in an arbitrarily large additive integer shift. We prove an exact inequality with factor $1/r$ for a power r, and give a criterion that combines it with an elliptic index-height estimate. Under rational *abc*, six fixed rational contacts and a quadratic input scale give the required strict height inequality; the precise absorption coefficient is $8/9$ for an explicit choice of exponent. A cubic input scale also gives a height criterion for five contacts. Four contacts behave differently: an explicit elliptic curve supplies rational arguments of unbounded height whose odd contact radical is one outside $\{2, 3, 5\}$. This last statement is unconditional. All point arithmetic and displayed polynomial identities in the example have an exact integer and rational certificate.

1 Heights and contact radicals

For a reduced fraction a/b, with $b > 0$, put

$$h(a/b) = \log \max(|a|, b).$$

For $u, v \in \mathbb{Q}$, write $u = A/D$, $v = C/D$ with $D > 0$ and $\gcd(D, A, C) = 1$, and define

$$h(1 : u : v) = \log \max(D, |A|, |C|).$$

Both $h(u)$ and $h(v)$ are at most this projective height. For a nonzero integer m, let $\mathrm{rad}(m)$ be the product of its distinct prime divisors, and let $\mathrm{sf}(m)$ be the product of primes occurring to odd exponent. Signs are omitted. For a prime p, the valuation v_p is normalized by $v_p(p) = 1$.

Let $\Theta \subset \mathbb{Q}$ be a finite set of distinct contacts. Choose a finite set S of primes containing the denominators of the contacts and all primes at which two contacts have the same reduction. For $t \notin \Theta$, put

$$M_{\Theta,S}(t) = \prod_{\substack{p \notin S:\ \text{some } \theta \in \Theta \text{ has} \\ v_p(t-\theta) > 0 \text{ odd}}} p. \tag{1}$$

The empty product is one. Enlarging S changes a logarithmic bound for this radical by at most a fixed additive constant.

We use the following form of rational *abc*: for every $\epsilon > 0$ there is $K_\epsilon \geq 1$ such that

$$c \leq K_\epsilon \,\mathrm{rad}(abc)^{1+\epsilon} \quad (a + b = c, a, b, c > 0, \gcd(a, b) = 1).$$

When a single exponent $\kappa > 1$ is sufficient, we state that we assume a constant $K_\kappa \geq 1$ with

$$c \leq K_\kappa \,\mathrm{rad}(abc)^{\kappa} \quad (a + b = c, a, b, c > 0, \gcd(a, b) = 1). \tag{2}$$

2 A height scale uniform in its additive shift

Lemma 2.1. *Let $r \geq 1$ and $B \geq 1$ be integers. Suppose that $\gamma \in \mathbb{Q}$ satisfies either $\gamma \geq 1$ or that r is even. Set*

$$\beta = \gamma^r + B.$$

Then $\beta \geq B$ and

$$h(\gamma) \leq \frac{1}{r} h(\beta). \tag{3}$$

In particular the inequality is uniform in B.

Proof. Write $\gamma = a/b$ with $\gcd(a, b) = 1$ and $b > 0$. The displayed fraction for β is already reduced, because

$$\gcd(a^r + Bb^r, b^r) = 1.$$

Under either stated condition, the numerator is positive and

$$a^r + Bb^r \geq \max(|a|^r, b^r).$$

It is also at least the denominator. Consequently

$$h(\beta) = \log(a^r + Bb^r) \geq r \log \max(|a|, b) = r h(\gamma).$$

□

Two input choices will be useful. The direct choice $\gamma = X$ permits every rational X when r is even. The positive choice

$$\gamma = 2X^2 + 1 \tag{4}$$

permits every $r \geq 1$. For rational X, it satisfies

$$\gamma \in \mathbb{Z} \quad \Longleftrightarrow \quad X \in \mathbb{Z}. \tag{5}$$

Indeed, if $X = c/d$ is reduced, integrality of $2X^2 + 1$ gives $d^2 \mid 2$, hence $d = 1$. For integer input, γ is a positive odd integer.

Theorem 2.2 (A uniform bound from a smaller input height)**.** *Let $T : \mathbb{Z} \setminus \{0\} \to \mathbb{Q}$ satisfy*

$$n^2 \leq A h(T(n)) + J \tag{6}$$

for fixed $A > 0$ and $J \geq 0$. Let $r \geq 1$, let $B \geq 1$ be an integer, and put $\beta = \gamma^r + B$ under the hypotheses of Lemma 2.1. Set

$$H = h(1 : \beta : T(n)).$$

Suppose that, whenever $\gamma \neq n$, the arithmetic conditions under consideration imply

$$H \leq \kappa \log N_\gamma + C, \qquad 0 < \kappa < r, \tag{7}$$

where N_γ is the positive absolute reduced numerator of $\gamma - n$ and $C \geq 0$ is independent of B, γ, n. Then $|n|$ and H are bounded independently of B whenever $\gamma \neq n$. In particular, for every sufficiently large B, these conditions imply $\gamma = n$.

Proof. If $\gamma = a/b$ is reduced with $b > 0$, then $\gcd(a - nb, b) = 1$ and

$$N_\gamma = |a - nb| \leq e^{h(\gamma)}(1 + |n|).$$

By Lemma 2.1,

$$\log N_\gamma \leq H/r + \log(1 + |n|).$$

Insert this into (7) and move the term $(\kappa/r)H$ to the left. With

$$c = \frac{\kappa}{1-\kappa/r}, \qquad C' = \frac{C}{1-\kappa/r},$$

one obtains

$$H \le c\log(1+|n|) + C'. \tag{8}$$

Since $h(T(n)) \le H$, equation (6) gives

$$n^2 \le Ac\log(1+|n|) + AC' + J \le Ac|n| + AC' + J.$$

Thus any integer $M > Ac + \sqrt{AC' + J}$ bounds $|n|$ strictly. It follows that

$$H \le c\log(1+M) + C' =: H_*.$$

But $H \ge h(\beta) \ge \log\beta \ge \log B$. Choosing $B > e^{H_*}$ rules out $\gamma \ne n$. □

Remark 2.3. For the positive input (4), the conclusion $\gamma = n$ proves $X \in \mathbb{Z}$ by (5); the completeness label $n = 2X^2 + 1$ never vanishes. For the direct input $\gamma = X$, the conclusion is already $X \in \mathbb{Z}$. If a polynomial $Q(X, \boldsymbol{Y})$ describes precisely the nonzero integers, then $XQ(X, \boldsymbol{Y})$ describes all integers, with the same variables and one higher degree. This last assertion is the elementary equivalence $XQ = 0 \iff X = 0$ or $Q = 0$ over $\mathbb{Q}$.

3 Contact and power-sum height estimates

Proposition 3.1. *Assume rational abc. Let Θ consist of $d \ge 5$ distinct rational contacts, and let S be as in (1). For every ϵ with $0 < 2\epsilon < d-4$, there is a constant $C_{\Theta,S,\epsilon} \ge 0$ such that*

$$h(t) \le \frac{1}{d-4-2\epsilon}\log M_{\Theta,S}(t) + C_{\Theta,S,\epsilon} \qquad (t \in \mathbb{Q}\setminus\Theta). \tag{9}$$

Proof. Write each contact as a_i/b_i in lowest terms with $b_i > 0$, and set

$$G(U,V) = \prod_{i=1}^{d}(b_iU - a_iV).$$

This integral homogeneous form has no repeated factor. For a primitive integer pair (u,v) with $v > 0$, put $H_0 = \max(|u|, v)$. Granville's consequence of rational *abc* [1, Proposition 1] gives

$$\log\operatorname{rad}(G(u,v)) \ge (d-2-\epsilon)\log H_0 - C_1.$$

The finitely many bounded heights can be included by enlarging C_1. Also $|G(u,v)| \le C_2 H_0^d$. The primewise inequality

$$\operatorname{rad}(m)^2 \le |m|\operatorname{sf}(m)$$

therefore gives

$$\log\operatorname{sf}(G(u,v)) \ge (d-4-2\epsilon)\log H_0 - C_3.$$

Outside S, at most one primitive linear factor can be divisible by a given prime, and a prime dividing v divides none. Its odd exponent occurs exactly at a positive odd valuation of one of $u/v - a_i/b_i$. Hence

$$\operatorname{sf}(G(u,v)) \le \left(\prod_{p\in S} p\right) M_{\Theta,S}(u/v).$$

Dividing the resulting inequality by $d-4-2\epsilon > 0$ proves (9). □

Lemma 3.2. *Assume (2) for a fixed $\kappa > 1$. Let $m \geq 4$ be even and put $\alpha = m/\kappa - 3$. If $\alpha > 0$, then*

$$h(1 : u : v) \leq \alpha^{-1} h(u^m + v^m) + C_{m,\kappa} \qquad (u, v \in \mathbb{Q}), \tag{10}$$

where one may take

$$C_{m,\kappa} = \frac{\alpha + 1}{m\alpha} \log 2 + \frac{1}{\kappa\alpha} \log K_\kappa.$$

Proof. If a coordinate is zero, the left side is $h(u^m + v^m)/m$; if both are zero it is zero. These cases satisfy the assertion. Otherwise write

$$u = gA/D, \qquad v = gC/D, \qquad \gcd(A, C) = \gcd(g, D) = 1, \quad g, D > 0.$$

Put $H_0 = \max(|A|, |C|)$ and $N = A^m + C^m$, so $H_0^m \leq N \leq 2H_0^m$. Set

$$k = \gcd(N, D^m), \qquad m_0 = N/k, \qquad n_0 = D^m/k, \qquad M = \max(g^m m_0, n_0).$$

Then $u^m + v^m = g^m m_0/n_0$ is reduced. Every prime dividing N either divides D or survives in m_0, and so $\operatorname{rad}(N) \leq D m_0$. Applying (2) to the pairwise coprime positive integers A^m, C^m, N gives

$$H_0^{m/\kappa - 2} \leq K_\kappa^{1/\kappa} D m_0.$$

Since $D^m = n_0 N/m_0 \leq 2n_0 H_0^m/m_0$, this implies

$$H_0^\alpha \leq 2^{1/m} K_\kappa^{1/\kappa} m_0^{(m-1)/m} n_0^{1/m}.$$

Moreover

$$\max(gH_0, D) \leq 2^{1/m} H_0 (M/m_0)^{1/m}.$$

Combining the logarithms of these inequalities gives

$$h(1 : u : v) \leq C_{m,\kappa} + \frac{m - 1 - \alpha}{m\alpha} \log m_0 + \frac{1}{m\alpha} \log n_0 + \frac{1}{m} \log M.$$

All three coefficients are nonnegative and sum to $1/\alpha$. Since $m_0, n_0 \leq M$, this proves (10). □

Rational *abc* supplies $\kappa = 16/15$. With $m = 4$, the lemma gives

$$h(1 : u : v) \leq \frac{4}{3} h(u^4 + v^4) + C_4, \qquad C_4 = \frac{7}{12} \log 2 + \frac{5}{4} \log K_{16/15}. \tag{11}$$

4 Six contacts and a quadratic input scale

Theorem 4.1. *Assume rational abc and the index-height bound (6). Fix six distinct rational contacts and the associated radical $M = M_{\Theta,S}$. Put*

$$\gamma = 2X^2 + 1, \qquad \beta = \gamma^2 + B, \qquad a_1 = \beta^4 + T(n)^4,$$

where $B \geq 3$ is an odd fixed integer and $n \neq 0$ is an integer label. Let $\lambda \in \mathbb{Q}^\times$. Suppose that, when $\gamma \neq n$, the local conditions give

$$M(\lambda) \mid N_\gamma, \qquad M(a_1\lambda) \mid N_\gamma, \tag{12}$$

with both arguments outside the contact set. Then every sufficiently large odd B forces $X \in \mathbb{Z}$. The sufficient bound is uniform in the input and witnesses.

Proof. In Proposition 3.1, take $d = 6$ and $\epsilon = 1/4$. Then

$$h(t) \le \frac{2}{3} \log M(t) + C_0.$$

By (12), the quotient-height inequality and (11) give, with $H = h(1 : \beta : T(n))$,

$$\begin{aligned} H &\le \frac{4}{3} h(a_1) + C_4 \\ &\le \frac{4}{3} (h(\lambda) + h(a_1 \lambda)) + C_4 \\ &\le \frac{16}{9} \log N_\gamma + C_2, \qquad C_2 = \frac{8}{3} C_0 + C_4. \end{aligned}$$

Lemma 2.1 yields $h(\gamma) \le H/2$. Consequently

$$H \le \frac{8}{9} H + \frac{16}{9} \log(1 + |n|) + C_2, \qquad H \le 16 \log(1 + |n|) + 9C_2.$$

Theorem 2.2, with $r = 2$ and $\kappa = 16/9$, now gives $\gamma = n$ for sufficiently large B. Equation (5) proves the assertion. □

The same calculation gives other height criteria. The table records choices in Proposition 3.1, using the fourth-power estimate (11). The column headed κ/r is the coefficient of H that is moved to the left in the proof of Theorem 2.2.

Contacts d	ϵ	Contact coefficient	Input power r	κ/r
8	1/2	1/3	1	8/9
6	1/4	2/3	2	8/9
5	1/36	18/17	3	16/17

For example, five contacts and $\beta = \gamma^3 + B$ give

$$H \le \frac{48}{17} \log N_\gamma + C \le \frac{16}{17} H + \frac{48}{17} \log(1 + |n|) + C,$$

and hence $H \le 48 \log(1 + |n|) + 17C$. These assertions are height implications under the displayed local divisibilities. A polynomial construction must separately supply those divisibilities and prove completeness of its local and pole conditions.

Remark 4.2. The smaller input height helps only when the comparison residual is $\gamma - n$. Keeping the residual $\beta - n$ gives $\log N_\beta \le h(\beta) + \log(1 + |n|)$, which does not acquire the factor $1/r$ from Lemma 2.1.

5 Four contacts with bounded odd radical

The degree restriction in Proposition 3.1 reflects a genuine obstruction. For four contacts its radical argument has exponent -2ϵ after the square part is removed. The following explicit example shows that no replacement logarithmic estimate can hold for this contact set.

Theorem 5.1. *Let*

$$\Theta = \{-1, 0, 2, 3\}, \qquad S \supseteq \{2, 3, 5\}$$

with S finite. There are infinitely many $t \in \mathbb{Q} \setminus (\Theta \cup \{1\})$ for which

$$M_{\Theta,S}(t) = 1.$$

Their heights are unbounded. Consequently there are no constants $c, C \ge 0$ such that

$$h(t) \le c \log M_{\Theta,S}(t) + C$$

for every rational t outside the contacts and the pole. No conjecture is used.

Proof. Consider the smooth quartic curve

$$10y^2 = (t+1)t(t-2)(t-3). \tag{13}$$

Its four roots are distinct, and it has the rational point $(t, y) = (4, 2)$. On $t \neq 0$, make the substitutions

$$X_E = \frac{60}{t}, \qquad Y_E = \frac{600y}{t^2}.$$

They give the elliptic curve

$$E: \quad Y_E^2 = X_E^3 + 10X_E^2 - 2400X_E + 36000 = (X_E + 60)(X_E - 20)(X_E - 30), \tag{14}$$

with inverse formulas

$$t = \frac{60}{X_E}, \qquad y = \frac{6Y_E}{X_E^2}.$$

The point $(4, 2)$ corresponds to $P = (15, 75)$. The tangent slope at P is $-19/2$, and the group law gives

$$2P = \left(\frac{201}{4}, \frac{2079}{8}\right). \tag{15}$$

To apply the integral short-form version of Nagell–Lutz, put

$$x = 9X_E + 30, \qquad Y = 27Y_E.$$

Then

$$Y^2 = x^3 - 197100x + 32130000. \tag{16}$$

This integral equation is nonsingular; its cubic has the distinct roots $-510, 210, 300$. The image of $2P$ has abscissa $1929/4$, which is not integral. Nagell–Lutz states that every rational torsion point on an integral short Weierstrass equation has integral coordinates; see [2, Theorem 1.1] or [3, Chapter II, Theorem 5.1]. Thus P has infinite order.

For each nonzero integer n, the point $[n]P$ is nonidentity and is not a point of order two. It cannot have $X_E = 0$, since $Y_E^2 = 36000$ has no rational solution. The inverse formulas therefore give a rational point (t_n, y_n) on (13) with $t_n \notin \Theta$. Also $t_n \neq 1$, since at $t = 1$ the equation would require $y^2 = 2/5$. There are infinitely many distinct t_n: a fixed value of t determines at most two points of (14), whereas the multiples of P are all distinct.

Fix $p \notin \{2, 3, 5\}$. If $v_p(t_n) < 0$, then all four differences $t_n - \theta$ have negative valuation, so p does not occur in (1). If $v_p(t_n) \geq 0$, at most one difference has positive valuation, since the nonzero differences among the contacts have prime factors only two and three. Equation (13) gives

$$v_p((t_n + 1)t_n(t_n - 2)(t_n - 3)) = 2v_p(y_n),$$

which is even. Therefore no positive contact valuation at such a prime is odd. It follows that $M_{\Theta,S}(t_n) = 1$.

Finally, only finitely many reduced rational fractions have height at most any fixed constant. Since the t_n form an infinite set, their heights are unbounded. This proves all assertions. □

Corollary 5.2. *The counterexamples in Theorem 5.1 may be chosen with $3 < t < 5$, and still have unbounded height.*

Proof. The real group $E(\mathbb{R})$ is compact. The closure of $\mathbb{Z}P$ is an infinite compact subgroup, hence has no isolated point: otherwise it would be discrete and compact, and therefore finite. Every sufficiently small neighborhood of $P = (15, 75)$ consequently contains infinitely many multiples of P. The rational function $60/X_E$ is continuous there and equals four at P. Choose the neighborhood so that its values lie in $(3, 5)$. The preceding finite-fiber and height arguments apply to this infinite subset. □

The theorem concerns a proposed height bound for the odd radical. It does not assert that these arguments extend to complete nonintegral witnesses of any Diophantine construction.

6 Exact certificate

The standard-library program `computation/contact_heights/check_input_height_amplification.py` checks the rational point and duplication in (14), the birational substitution identity, the integral change to (16), and the nonintegral doubled abscissa. It also verifies the distinct roots, the excluded rational values, and the exact rational constants in the height table. Its machine-readable result is `input-height-amplification-certificate.json` in the same directory. The infinitude assertion follows from the proved nontorsion criterion and the argument above; it is not inferred from a finite search over multiples.

References

[1] A. Granville, Rational and integral points on quadratic twists of a given hyperelliptic curve, *International Mathematics Research Notices* (2007), article rnm027, Proposition 1, https://doi.org/10.1093/imrn/rnm027; author manuscript, https://dms.umontreal.ca/~andrew/PDF/hyperelliptics.pdf.

[2] L. Alpoge, *Nagell–Lutz, quickly*, author manuscript, Theorem 1.1, https://people.math.harvard.edu/~alpoge/papers/nagell-lutz,%20quickly.pdf.

[3] J. S. Milne, *Elliptic Curves*, second edition, World Scientific, 2021, Chapter II, Section 5, https://www.jmilne.org/math/Books/EC2.pdf.